Twisted cohomology on affine varieties

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March 2024

1 Basic set up

We start with a smooth affine variety X over an algebraically closed field k of characteristic zero. Its coordinate ring will be denoted as

$$k[X] = \frac{k[x_1, ..., x_2]}{\langle h_1, ..., h_m \rangle}$$

where the polynomials $h_1, ..., h_m \in k[x_1, ..., x_2]$ defines $X \in \mathbb{A}^n_k$ as their zero set.
The Kähler differentials are then

$$\Omega^1(X) = \frac{\bigoplus_{i=1}^n k[X] dx^i}{\langle dh_1, ..., dh_m \rangle} \tag{1}$$

which are the algebraic analogues of differential 1-forms in the case of algebraic varieties. The higher forms are then constructed by simply taking wedge products (which is an anti commuting algebra)

$$\Omega^k(X) = \bigwedge^k \Omega^1(X) \tag{2}$$

The crucial point to remember is that the module $\Omega^1(X)$ of Kähler differentials is a free module of rank n where n is precisely the dimension of the variety X. Everything constructed thus far are the same as was constructed in the beginning of chapter 4, but here's the modification we perform when we define the differential differently than was defined in (4.5)

$$d_\theta : \Omega^k(X) \to \Omega^{k+1}(X) \tag{3}$$

acting explicitly as

$$d_\theta(\alpha) = d\alpha + \theta \wedge \alpha \tag{4}$$

for a k-form $\alpha \in \Omega^k(X)$. The exterior derivative has been "twisted" with a closed 1-form $\theta \in \Omega^1(X)$. We wish to compute cohomology from the complex

$$0 \to \Omega^0_{X/k} \overset{d^0_\theta}{\to} \Omega^1_{X/k} \overset{d^1_\theta}{\to} \Omega^2_{X/k} \overset{d^2_\theta}{\to} .... \overset{d^{n-1}_\theta}{\to} \Omega^n_{X/k} \overset{d^n_\theta}{\to} 0 \tag{5}$$

that looks like

$$H^i_\theta(X/k) := \frac{Ker(d^i_\theta : \Omega^i_{X/k} \to \Omega^{i+1}_{X/k})}{Im(d^{i-1}_\theta : \Omega^{i-1}_{X/k} \to \Omega^i_{X/k})} \tag{6}$$

2 Examples

Let us look at an easier case: a smooth plane curve given by the zero set of some polynomial $P(x, y) \in k[x, y]$ on the affine 2-plane $\mathbb{A}^2_k$ that we denote by $\mathcal{C}$. Pick a Kähler 1-form $\alpha \in \Omega^1(\mathcal{C})$ which has the form

$$\alpha = g(x, y)dx + h(x, y)dy$$

where $g, h \in k[\mathcal{C}]$. We want to know what the nontrivial kernel is for a map like

$$d^i_\theta : \Omega^i_{\mathcal{C}/k} \to \Omega^{i+1}_{\mathcal{C}/k} \tag{7}$$

at the i-th piece of the complex

$$0 \to \Omega^0_{\mathcal{C}/k} \overset{d^0_\theta}{\to} \Omega^1_{\mathcal{C}/k} \overset{d^1_\theta}{\to} 0 \tag{8}$$

where $i = 0, 1$ and what the closed 1-form should look like so the kernel of the twisted derivative remains nontrivial in most of the pieces. As the curve as dimension one, its doesn't have nontrivial two forms so $\Omega^2_{\mathcal{C}/k} = 0$ and hence $Im(d^1_\theta) = 0$ and the whole of $\Omega^1_{\mathcal{C}/k}$ is mapped to zero, so the kernel $Ker(d^1_\theta) = \Omega^1_{\mathcal{C}/k}$. So to find $H^1_\theta(\mathcal{C}/k)$ we need the image $Im(d^0_\theta)$.
Let us take the circle for example so that $P(x, y) = x^2 + y^2 - r^2$. Then the coordinate ring of $\mathcal{C}$ is,

$$k[\mathcal{C}] = \frac{k[x, y]}{\langle x^2 + y^2 - r^2 \rangle} = \{f + y \cdot g \mid f, g \in k[x]\}$$
$$\therefore \Omega^0_{\mathcal{C}/k} = \{f + y \cdot g \mid f, g \in k[x]\}$$

So the kähler differentials $\Omega^1_{\mathcal{C}/k}$ is,

$$\Omega^1_{\mathcal{C}/k} = \frac{k[\mathcal{C}]dx + k[\mathcal{C}]dy}{\langle xdx + ydy \rangle}$$

Consider the polynomials $r^2 - x^2, -x \in k[x]$. They are co-prime in $k[x]$ and $k[x]$ is PID. Therefore, we have $f, g \in k[x]$ such that $f \cdot \left(r^2 - x^2\right) - g \cdot x \equiv 1$. Now consider $w = yfdx + gdy$ We claim that $[\omega]$ generate $\Omega^1_{\mathcal{C}/k}$. To prove this, it is sufficient to show that ω generates both dx and dy under the relation $xdx + ydy = 0$.

$$\begin{aligned}
& y\omega \\
&= y(yfdx + gdy) \\
&= y^2 fdx + ygdy \\
&= \left(r^2 - x^2\right) fdx + g \cdot (-xdx) \\
&= \left[\left(r^2 - x^2\right) f - xg\right] dx \\
&= 1 \cdot dx
\end{aligned}$$

So $dx = y\omega$ Similarly

$$\begin{aligned}
& - x\omega \\
&= -x(yfdx + gdy) \\
&= -yfxdx - gxdy \\
&= yf \cdot ydy - gxdy \\
&= ((r^2 - x^2)f - gx)dy \\
&= 1 \cdot dy
\end{aligned}$$

We have $dy = -x\omega$. Therefore ω generate both dx and dy. Now from the above information, we can deduce that,

$$\Omega^1_{\mathcal{C}/k} = R[\mathcal{C}] \cdot \omega = \{(f + g \cdot y)\omega \mid f, g \in k[x]\}$$
$$\begin{aligned}
\Omega^2_{\mathcal{C}/k} &= \Omega^1_{\mathcal{C}/k} \wedge \Omega^1_{\mathcal{C}/k} \\
&= \{(C_1 + D_1 y)\, \omega \wedge (C_2 + D_2 y)\, \omega \mid C_1, C_2, D_1, D_2 \in k[x]\}
\end{aligned}$$

Since, $w \wedge w$
$= (yfdx + gdy) \wedge (yfdx + gdy)$
$= yfdx \wedge yfdx + yfdx \wedge gdy + gdy \wedge yfdx + g^2 dy \wedge dy$
$= 0 + yfgdx \wedge dy + yfydy \wedge dx + 0$
$= yfgdx \wedge dy - yfgdx \wedge dy$
$= 0$

Therefore $\Omega^2_{\mathcal{C}/k} = 0$. So the twisted complex is:

$$0 \to \Omega^0_{\mathcal{C}/k} \xrightarrow{d^0_\theta} \Omega^1_{\mathcal{C}/k} \xrightarrow{d^1_\theta} 0$$

Let us fix a closed algebraic one form

$$\theta = xdy + ydx \tag{9}$$

Here the kernel of the map d^1_θ is the whole module $\Omega^1_{\mathcal{C}/k}$. To find $H^1_\theta(\mathcal{C}/k)$ we need the image $Im(d^0_\theta)$. Pick $f + y.g \in \Omega^0_{\mathcal{C}/k}$ for which we have

$$\begin{aligned}
&d_\theta(f + y.g) = d(f + y.g) + \theta \wedge (f + y.g)\\
&= d(f + y.g) + (xdy + ydx) \wedge (f + y.g)\\
&= f_{,x}\, dx + g_{,x}\, ydx + gdy + fxdy + xygdy + fydx + y^2 gdx\\
&= f_{,x}\, dx + g_{,x}\, ydx + gdy + fxdy + xygdy + fydx + (r^2 - x^2)gdx\\
&= [f_{,x} + g_{,x}\, y + fy + (r^2 - x^2)g]dx + (g + fx + xyg)dy\\
&= [f_{,x} + g_{,x}\, y + fy + (r^2 - x^2)g - x^2 g]dx + (g + fx)dy\\
&= [P(x) + yQ(x)]dx + H(x)dy
\end{aligned}$$

where $P(x) = f_{,x} + (r^2 - x^2)g - x^2 g$,$Q(x) = f + g_{,x}$ and $H(x) = fx + g$.
The question that arises, is that whether we can always find two polynomials $f_{,x}$ and $g(x) \in k[x]$ such that given any two polynomials $P(x), Q(x) \in k[x]$ the following two relations hold

$$P(x) = f_{,x} + (r^2 - 2x^2)g \tag{10}$$

$$Q(x) = f + g_{,x} \tag{11}$$

Let us first look at the case of (10). Assume that the $P(x)$ is a degree n polynomial which we write as

$$P(x) = p_n x^n + p_{n-1}x^{n-1} + p_{n-2}x^{n-2} + ... + p_2 x^2 + p_1 x + p_0$$

So

$$\begin{aligned}
f_{,x}\,(x) &= f_n x^n + f_{n-1}x^{n-1} + f_{n-2}x^{n-2} + ... + f_2 x^2 + f_1 x + f_0\\
g(x) &= g_{n-2}x^{n-2} + g_{n-3}x^{n-3} + g_{n-4}x^{n-4} + ... + g_2 x^2 + g_1 x + g_0
\end{aligned}$$

Comparing the coefficients using (10) we get

$$\begin{aligned}
p_n &= f_n - 2g_{n-2}\\
p_{n-1} &= f_{n-1} - 2g_{n-3}\\
p_{n-2} &= f_{n-2} + r^2 g_{n-2} - 2g_{n-4}\\
p_{n-3} &= f_{n-3} + r^2 g_{n-3} - 2g_{n-5}\\
&\vdots\\
p_2 &= f_2 + r^2 g_2 - 2g_0\\
p_1 &= f_1 + r^2 g_1\\
p_0 &= f_0
\end{aligned}$$

So the answer to question that we imposed above regarding (10) seems to affirmative and obviously its affirmative for (11) as well.

Using the relations

$$dy = -x\omega$$

$$dx = y\omega$$

that was shown above, we see that

$$\begin{aligned}
&d_\theta(f + y.g) = ([P(x) + yQ(x)]y - H(x)x)\omega\\
&= ([P(x)y + y^2 Q(x)] - H(x)x)\omega\\
&= (P(x)y + (r^2 - x^2)Q(x) - H(x)x)\omega\\
&= (P(x)y + W(x))\omega
\end{aligned}$$

here

$$\begin{aligned}
W(x) &= (r^2 - x^2)Q(x) - xH(x)\\
&= (r^2 - x^2)(f + g_{,x}) - x(fx + g)\\
&= r^2 f + r^2 g_{,x} - 2x^2 f - x^2 g_{,x} - gx
\end{aligned}$$

Let us assume again that $W(x)$ is an $n+3$ degree polynomial that might have the form

$$W(x) = w_{n+3}x^{n+3} + w_{n+2}x^{n+2} + w_{n+1}x^{n+1} + ... + w_2x^2 + w_1x + w_0$$

Here

$$g_{,x}(x) = g_{n-2}(n-2)x^{n-3} + g_{n-3}(n-3)x^{n-4} + g_{n-4}(n-4)x^{n-5} + ... + 3g_3x^2 + 2g_2x + g_1$$

$$\begin{aligned} f(x) &= \text{"}\int\text{"} f_{,x}(x)dx \\ &= \frac{f_n}{n}x^{n+1} + \frac{f_{n-1}}{n-1}x^n + \frac{f_{n-2}}{n-2}x^{n-1} + ... + \frac{f_2}{2}x^3 + \frac{f_1}{1}x^2 + f_0x + c \\ &= \overline{f_{n+1}}x^{n+1} + \overline{f_n}x^n + \overline{f_{n-1}}x^{n-1} + ... + \overline{f_3}x^3 + \overline{f_2}x^2 + \overline{f_1}x + \overline{f_0} \end{aligned}$$

(where $\overline{f_m} = \frac{f_{m-1}}{m-1}$ for $m > 1$ and $c = \overline{f_0}$) Now comparing the coefficients

$$\begin{aligned} w_{n+3} &= -2\overline{f_{n+1}} \\ w_{n+2} &= -2\overline{f_n} \\ w_{n+1} &= -2\overline{f_{n-1}} \\ w_n &= r^2\overline{f_n} - 2\overline{f_{n-2}} \\ w_{n-1} &= r^2\overline{f_{n-1}} - (n+1)g_{n-2} \\ w_{n-2} &= r^2\overline{f_{n-2}} - 2\overline{f_{n-4}} - (n+2)g_{n-3} \\ w_{n-3} &= r^2\overline{f_{n-3}} - 2\overline{f_{n-5}} + r^2(n-2)g_{n-2} - (n+3)g_{n-4} \\ w_{n-4} &= r^2\overline{f_{n-4}} - 2\overline{f_{n-6}} + r^2(n-3)g_{n-3} - (n+4)g_{n-5} \end{aligned}$$

We see that the polynomials f, g that can generate $P(x)$, cannot generate $W(x)$ of just any degree, it has to be that the degree of $W(x)$ must be 3 higher than the degree of $P(x)$. So the first cohomology is then

$$H^1_\theta(\mathcal{C}/k) =$$

3 Some properties

Over some algebraically closed field k of characteristic zero, we consider two smooth affine varieties X and Y as well as their product $X \times Y$. We also have the projection maps

$$pr_X : X \times Y \to X$$

$$pr_Y : X \times Y \to Y$$

We also consider Kähler differentials on them, namely

$$\psi_X \in \Omega^1_{X/k}$$

$$\psi_Y \in \Omega^1_{Y/k}$$

$$\psi \in \Omega^1_{X\times Y/k}$$

where $\psi = pr_X^*\psi_X + pr_Y^*\psi_Y$

Set the map

$$\Phi : H^l_{\psi_X}(X/k) \otimes_k H^m_{\psi_Y}(Y/k) \to H^{l+m}_{\psi}(X \times Y/k) \tag{12}$$

that acts as a cup product: take $[\alpha] \in H^l_{\psi_X}(X/k)$, $[\beta] \in H^l_{\psi_Y}(Y/k)$ where α is a d_{ψ_X}-closed l-form and β is a d_{ψ_Y}-closed m-form, the image of the map Φ, $\Phi([\alpha] \otimes [\beta])$ is a d_ψ-closed $(l+m)$-form.

4 Projective setting

For projective varieties over an algebraically closed field k of characteristic zero, we know that we cannot globally define Kähler differentials on it, so we cannot directly twist the algebraic de rham cohomology as we did in the affine case. So to define, we use the Mayer Vietoris sequence for algebraic de Rham cohomologies. Before giving the precise (and general) definition of the twisted algebraic de Rham cohomology for projective varieties, we first look at a concrete example, namely that of the projective line over the field k, $\mathbf{P}^1_k$.
The projective line has the affine covers

$$U_0 = \{[z_0 : z_1] : z_0 \neq 0\} \simeq \mathbf{A}^1_k = Spec(k[x])$$

$$U_1 = \{[z_0 : z_1] : z_1 \neq 0\} \simeq \mathbf{A}^1_k = Spec(k[\frac{1}{x}])$$

So the Kähler differentials on these affine open covers are

$$\Omega^1(U_0) = \{f(x)dx : f \in k[x]\}$$

$$\Omega^1(U_1) = \{f(x^{-1})d(x^{-1}) : f \in k[x^{-1}]\}$$
$$\Omega^1(U_0 \cap U_1) = \{f(x, x^{-1})dx : f \in k[x, x^{-1}]\}$$

Now we pick the twisting differential forms

$$\psi_0 = p(x)dx \in \Omega^1(U_0)$$

$$\psi_1 = p(x^{-1})d(x^{-1}) \in \Omega^1(U_1)$$
$$\psi_{0,1} = p(x, x^{-1})dx \in \Omega^1(U_0 \cap U_1)$$

The twisted de Rham cohomology for the projective line will be found in the Mayer Vietoris sequence

$$0 \to H^0_\psi(\mathbf{P}^1_k) \xrightarrow{i^0} H^0_{\psi_0}(U_0) \oplus H^0_{\psi_1}(U_1) \xrightarrow{j_0} H^0_{\psi_{0,1}}(U_0 \cap U_1) \xrightarrow{\delta} \tag{13}$$

$$H^1_\psi(\mathbf{P^1_k}) \xrightarrow{i^1} H^1_{\psi_0}(U_0) \oplus H^1_{\psi_1}(U_1) \xrightarrow{j_1} H^1_{\psi_{0,1}}(U_0 \cap U_1) \to 0$$